

Nonparametric survival analysis project

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Part B: Nonparametric Survival Analysis

In this part, the response variable is the survival time of a satellite, measured by `time_days`. The censoring indicator is `event`, where

$$\delta_i = \begin{cases} 1, & \text{if the satellite decayed,} \\ 0, & \text{if the satellite was censored.} \end{cases}$$

Thus the observed time is

$$Y_i = \min(T_i, C_i),$$

where T_i is the true lifetime of satellite i , and C_i is the censoring time.

The categorical variable used in this analysis is satellite purpose. To avoid having too many groups, the original categories were grouped into three categories:

Communications, Earth Observation, Other.

The group “Other” contains Navigation, Other, Science, and Technology satellites.

(a) Kaplan–Meier survival curves

For each purpose group, the survival function was estimated nonparametrically using the Kaplan–Meier estimator:

$$\hat{S}(t) = \prod_{j: Y_{(j)} \leq t} \frac{r(j) - d(j)}{r(j)}.$$

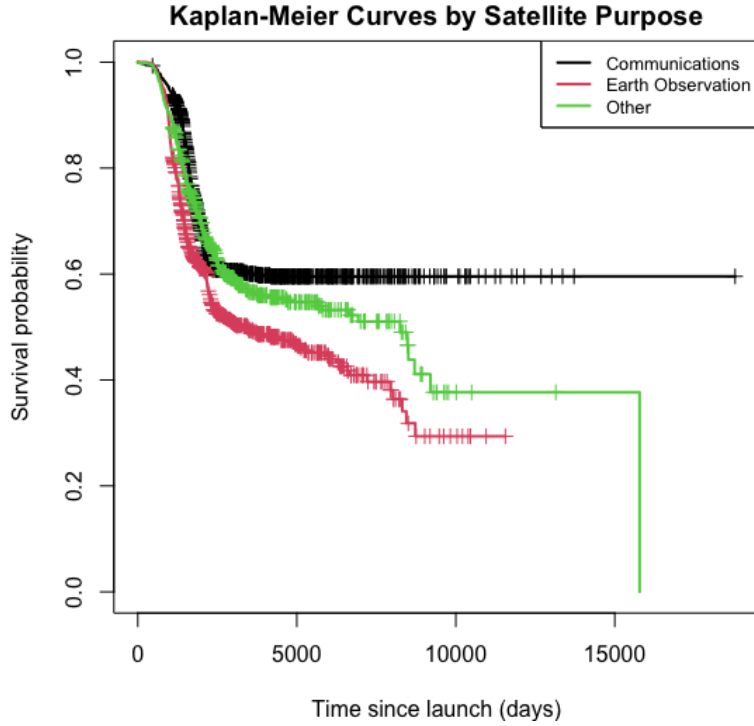


Figure 1: Kaplan–Meier survival curves by satellite purpose. Censoring times are indicated by tick marks.

Figure 1 shows clear differences between the three survival curves. The Earth Observation curve declines most rapidly, especially during the first few thousand days after launch. This suggests that Earth Observation satellites tend to experience decay earlier than the other groups.

The Communications curve remains above the other curves for almost the entire follow-up period and does not fall below 0.5. This indicates longer survival for Communications satellites and explains why the median survival time for this group could not be estimated. The large number of tick marks on this curve also shows substantial censoring, meaning many Communications satellites were still active at the end of follow-up.

The Other group initially behaves similarly to the Communications group but later declines more strongly. Its long right tail suggests that some satellites in this group survive for a very long time, but the late part of the curve should be interpreted cautiously because fewer satellites remain at risk and censoring becomes more influential.

Overall, the graph suggests that satellite survival differs by purpose. Earth Observation satellites appear to have the shortest survival, Communications satellites the longest survival, and the Other group lies between them for much of the observation period.

(b) Quartiles of the survival curves

The p -th quantile of the survival distribution is estimated by

$$\hat{x}_p = \inf\{t : \hat{S}(t) \leq 1 - p\}.$$

Therefore, the first quartile corresponds to the time at which the survival curve reaches 0.75, the median corresponds to the time at which the survival curve reaches 0.50, and the third quartile corresponds to the time at which the survival curve reaches 0.25.

The estimated quartiles and confidence intervals were:

Group	Q_1	L_{Q_1}	U_{Q_1}	Median	L_M	U_M	Q_3	L_{Q_3}	U_{Q_3}
Communications	1744	1689	1803	NA	NA	NA	NA	NA	NA
Earth Observation	1309	1249	1365	3438	2429	5028	NA	8446	NA
Other	1697	1511	1868	8249	5677	9205	15781	NA	NA

For Communications satellites, the first quartile is estimated as 1744 days, with a 95% confidence interval from 1689 to 1803 days. The median and third quartile are not estimable because the Kaplan–Meier curve does not fall below 0.50 or 0.25. This is due to heavy censoring and indicates that more than half of Communications satellites are still surviving by the end of follow-up.

For Earth Observation satellites, the first quartile is 1309 days, with a 95% confidence interval from 1249 to 1365 days. The median survival time is 3438 days, with a 95% confidence interval from 2429 to 5028 days. The third quartile is not fully estimable because the survival curve does not provide enough information in the lower tail.

For the Other group, the first quartile is 1697 days, with a 95% confidence interval from 1511 to 1868 days. The median is much larger, estimated as 8249 days, with a 95% confidence interval from 5677 to 9205 days. The third quartile is estimated as 15781 days, although its confidence interval is not fully estimable. This reflects uncertainty in the tail of the survival curve due to censoring.

Overall, the quartile estimates suggest that Earth Observation satellites tend to decay earlier than the other groups, while Communications satellites have the longest observed survival profile.

(c) Test for differences between survival curves

To compare the survival curves, a log-rank test was used. The null hypothesis is

$$H_0 : h_1(t) = h_2(t) = h_3(t) \quad \text{for all } t,$$

where $h_k(t)$ is the hazard function for group k . The alternative hypothesis is that at least one group has a different hazard function:

$$H_1 : h_i(t) \neq h_j(t) \quad \text{for at least one pair } i, j.$$

The log-rank test compares observed and expected numbers of events over time. For each event time, the test is based on differences of the form

$$O_j - E_j,$$

where O_j is the observed number of events and E_j is the expected number of events under the null hypothesis of equal hazards. For more than two groups, the resulting test statistic has approximately a chi-squared distribution with $l - 1$ degrees of freedom, where l is the number of groups.

The log-rank test gave:

$$\chi^2 = 85.4, \quad df = 2, \quad p < 2 \times 10^{-16}.$$

Since the p-value is extremely small, we reject H_0 . There is very strong evidence that the survival curves differ between the three satellite-purpose groups.

The observed and expected events were:

Group	Observed	Expected
Communications	1219	1387
Earth Observation	571	411
Other	284	276

Earth Observation satellites had more observed events than expected under equal survival, while Communications satellites had fewer observed events than expected. This supports the graphical conclusion that Earth Observation satellites have shorter survival and Communications satellites have longer survival.

As an additional check, a Wilcoxon-type test was also performed using $\rho = 1$, which gives more weight to earlier differences in survival. This test gave

$$\chi^2 = 104, \quad df = 2, \quad p < 2 \times 10^{-16}.$$

This also rejects the null hypothesis, showing that the difference between groups is already strong at earlier times.

Therefore, both the Kaplan–Meier curves and the hypothesis tests indicate that satellite survival differs significantly according to satellite purpose.